Faculty of Computing

SECP2523: DATABASE

SEM I 2025/ 2026

INITIAL PROPOSED SYSTEM

SYSTEM NAME : Hasta Vehicle Rental Booking System

Group Name: 404NotFound

Members:

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Stakeholder: Hasta Travel & Tours Sdn Bhd

Representative Name	Date Interviewed
Encik Alif Firdaus (Owner of Hasta Travel's & Tours Sdn Bhd)	31 October 2025, 2 November 2025, 13 November 2025, 4 December 2025, 11 December 2025, 6 January 2026.

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PROPOSED SYSTEM BACKGROUND

1. Introduction

Hasta Travel & Tours Sdn. Bhd. is a car rental company that operates as a car rental service provider and offers short-term vehicle rental service specifically for Universiti Teknologi Malaysia (UTM) students and staff. This proposal outlines the design and development of a centralized relational database system for Hasta Travel & Tours Sdn. Bhd.. Hasta Travel manages daily rental activities that involve managing vehicles ,verifying customers , processing payment, handling deposit, penalties and loyalty rewards. These operations generate a large volume of interrelated data that must be recorded accurately and consistently to ensure smooth service delivery.

However, the current data management approach used by Hasta Travel relies heavily on manual processes and disconnected digital tools, including WhatsApp, WhatsForm, Google Sheets, physical documents, whiteboards, files in computers and the Drive Wahdah system. These tools operate independently rather than as part of a one data platform, resulting in fragmented records, duplicated data entry, and a high dependency on staff intervention. The company adopts a file-based system approach, where information is stored in separate files on computers rather than within a centralized database. Important rental information such as booking details, payment status, and customer records is separated across multiple sources, making data difficult to retrieve and increasing the risk of inconsistency and human error.

The absence of a centralized database also leads to several operational challenges, including duplicate or conflicting bookings, weak payment record traceability, ineffective blacklist enforcement, and limited reporting capability. Staff are required to manually cross-check information across different platforms to verify vehicle availability, confirm payments, and monitor customer history.

In response to these challenges, this proposal suggests the implementation of a structured database system that centralizes all rental-related data into a single, authoritative source. An automated web-based car rental management system that combines all reservations, payments, and user data into a single, safe platform is desperately needed, as these problems demonstrate.The proposed database will support accurate record-keeping, reduce redundancy, and enable efficient querying and reporting. By replacing the existing fragmented data handling approach with a well-designed relational database, Hasta Travel will be better equipped to improve operational efficiency, enhance data reliability, and support informed decision-making for future growth.

2. The Existing System

2.1 Purpose and What the Existing System Does

The existing booking system used by Hasta Travel operates through manual processes and basic digital tools, resulting in several limitations in daily operations. The daily operations such as customer booking, payment verification and record keeping are handled using physical documents, digital forms and simple spreadsheets files. The current system for booking cars that is used in Hasta Travel is DRIVE WAHDAH. The system is only for client-side and only accessible by staff and admin of HASTA Travel & Tours Sdn Bhd.

2.2 Type of Existing System (Manual / Automated)

The existing system is mainly manual with limited digital support. The car rental operations rely on Whatsapp, WhatsForm, physical documents, manual boards, and spreadsheets, while Drive Wahdah is used by staff/admin to store booking and payment information.

2.3 Existing System Workflow / Processes

2.3.1 Car Rental Booking Process

2.3.1.1 Getting Information

The car rental booking process at Hasta Travel is handled through Whatsapp or Whatsform. Currently, customers send a booking request message to rent a car via Whatsapp. The staff will immediately provide a booking template that includes booking information such as a confirmation of whether the customer is a UTM student of staff, pickup date, pickup time, pickup location, return date, return time, return location and destination to the customers and customers must fill in before proceeding with the reservation. The template strictly informs the customer that the bookings require a minimum of two days for trips outside Johor.

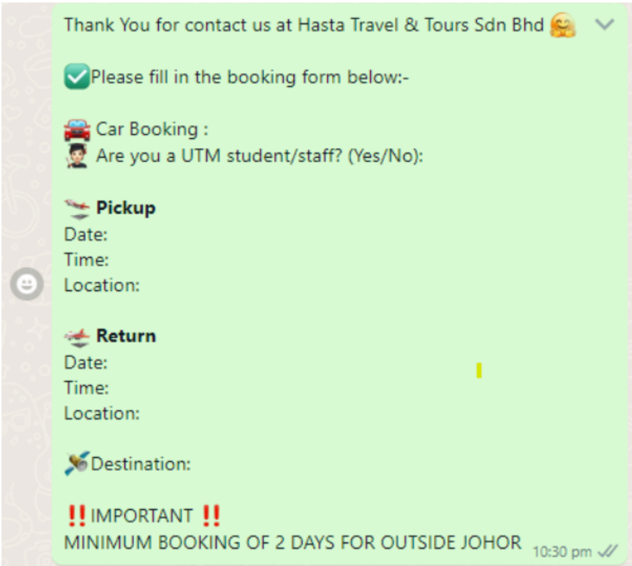


Figure 2.1: Booking Template

The staff of Hasta Travel will review the customer's reply to ensure that all booking details meet the requirements. They will verify that the customer is a UTM student or staff member, confirm that the pickup and return locations are at UTM Student Mall, and ensure that the pickup and return times fall within office hours. The booking is strictly limited to UTM students and staff. Any booking made by a non-UTM individual will be rejected. Hasta Travel currently uses a manual record system, such as a sales board, to display car availability for booking on specific dates .

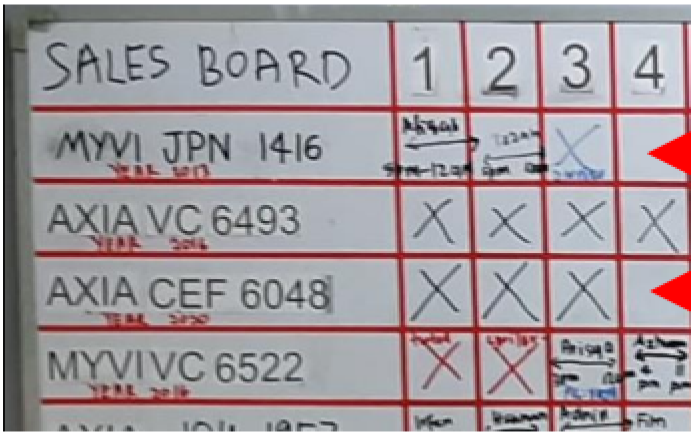
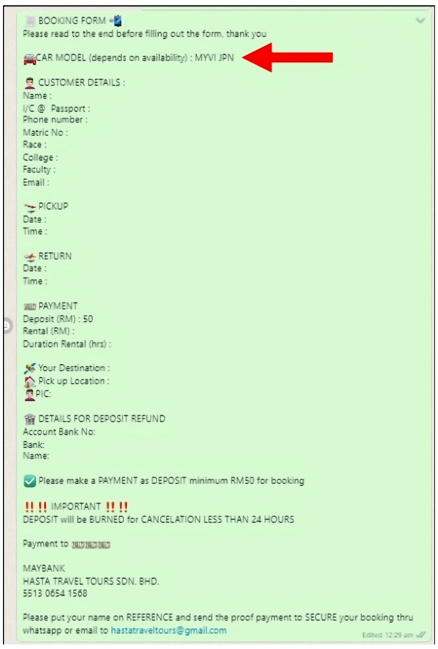


Figure 2.2: Sales Board Hasta Travel

If the certain dates marked with an "X", indicating the car are unavailable, which other slots remain open. This system allows staff to inform customers about available cars and confirm their preferred choice. The staff will inform customers that the price is different for different types of car.

The staff will provide the booking form to customers after confirming their chosen car .The booking forms will include customer details, pickup and return details, payment details , destination, details for deposit refund. The booking will inform the user that the deposit has a minimum RM50 for booking, deposit will be burned for cancellation less than 24 hours and Payment Reference ID.



BOOKING FORM
Please read to the end before filling out the form, thank you

CAR MODEL (depends on availability) : MYVI JPN

CUSTOMER DETAILS :
Name :
I/C @ Passport :
Phone number :
Matric No :
Race :
College :
Faculty :
Email :

PICKUP
Date :
Time :

RETURN
Date :
Time :

PAYMENT
Deposit (RM) : 50
Rental (RM) :
Duration Rental (hrs) :

Your Destination :
Pick up Location :
PIC:

DETAILS FOR DEPOSIT REFUND
Account Bank No:
Bank:
Name:

Please make a PAYMENT as DEPOSIT minimum RM50 for booking

‼ ‼ IMPORTANT ‼ ‼
DEPOSIT will be BURNED for CANCELATION LESS THAN 24 HOURS

Payment to

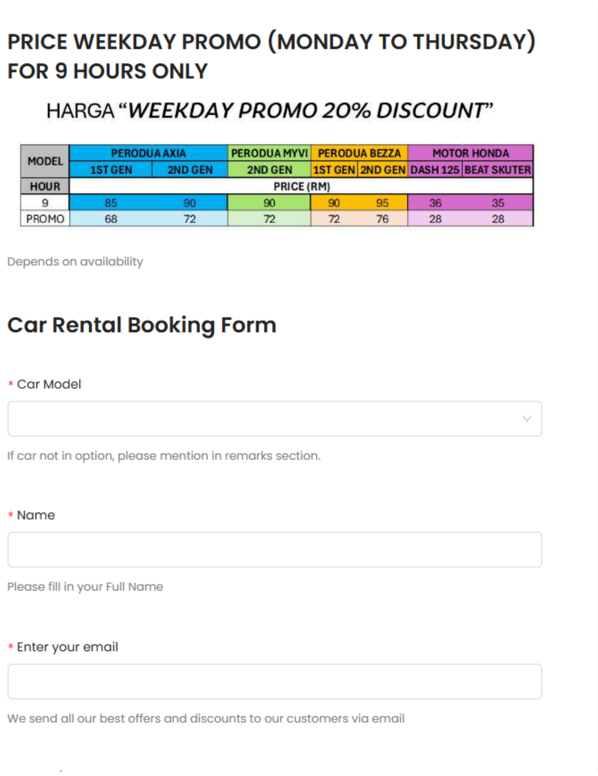
MAYBANK
HASTA TRAVEL TOURS SDN. BHD.
5513 0654 1568

Please put your name on REFERENCE and send the proof payment to SECURE your booking thru whatsapp or email to hastatraveltours@gmail.com

Figure 2.3: Booking Form

If the customer is not booking the car during staff operating hours and the location is not UTM Student Mall or after the customer pays the deposit, the staff will ask the customer to provide their pictures of IC or passport, driving license and matric card in a single PDF.

Another way for customers to book a car is filling in WhatsForm that was prepared by Hasta Travel. The WhatsForm allows customers to book a car by displaying the rental price per hour for each car type and available promotions on weekdays from Monday to Thursday for 9 hour bookings. It collects essential information for booking from customers, including desired car model, name, email, student ID or staff ID, IC number or passport number, phone number, address and faculty. It also collects pickup and return details such as pickup date, pickup time, pickup location, return date, return time and return location, rental duration, rental fee, destination and additional remarks such as car preferences, Bluetooth, promotions, promo codes or vouchers. Once completed, the WhatsForm automatically converts the information into a booking template and sends it directly to Hasta Travel's Whatsapp for processing.



PRICE WEEKDAY PROMO (MONDAY TO THURSDAY) FOR 9 HOURS ONLY

HARGA "*WEEKDAY PROMO 20% DISCOUNT*"

MODEL	PERODUA AXIA		PERODUA MYVI	PERODUA BEZZA		MOTOR HONDA	
	1ST GEN	2ND GEN	2ND GEN	1ST GEN	2ND GEN	DASH 125	BEAT SKUTER
HOUR	PRICE (RM)						
9	85	90	90	90	95	36	35
PROMO	68	72	72	72	76	28	28

Depends on availability

Car Rental Booking Form

* Car Model

If car not in option, please mention in remarks section.

* Name

Please fill in your Full Name

* Enter your email

We send all our best offers and discounts to our customers via email

Figure 2.4: WhatsForm for Booking Car

2.3.1.2 Deposit

The staff will remind the customer to fill in the Booking Form and pay the deposit. Then, the staff will ask for the receipt from the customer and verify that the RM50 deposit amount is correct. They will also ensure that the deposit time matches the time of the payment was received. If the customer forgets to pay or is late in paying the deposit, the staff of Hasta Travel will remind the customer to make the payment.If the customer asks about the deposit, the staff will explain that the RM50 deposit will be returned within 10 working days if the car has no damage, the oil level is the same as before, there are no summonses, the car is clean and it is returned on time. These checks allow the staff to refund the deposit without the deductions. After receiving the deposit from the customer, the staff of Hasta Travel will download the deposit receipt and save it in the specific folder for that year and specific file for that month. The file name format is <year, month, day> <RM50> <name>.

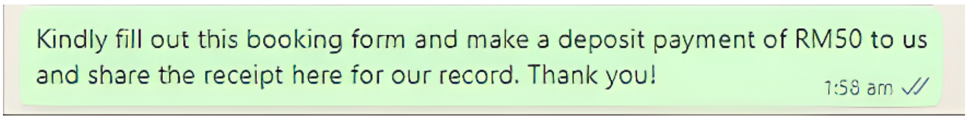


Figure : Reminder for customer to pay the deposit

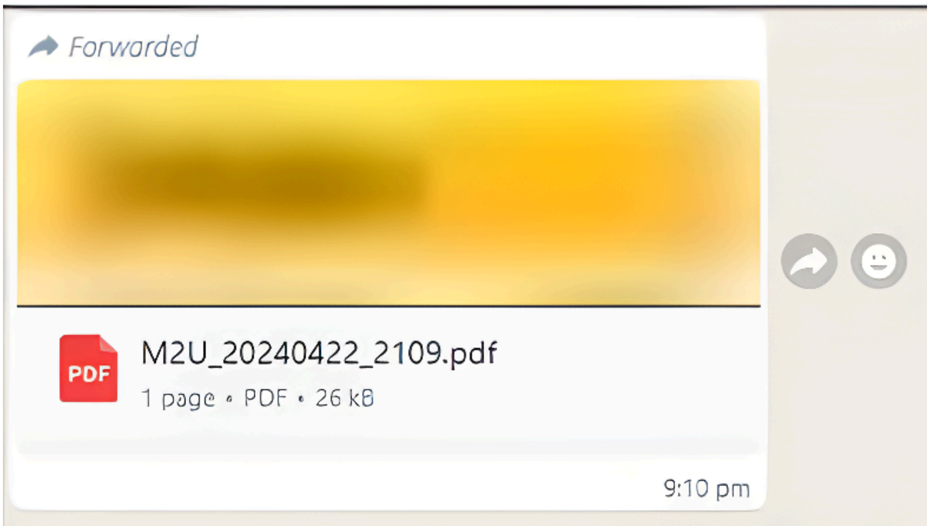


Figure 2.5: Receipt of Deposit

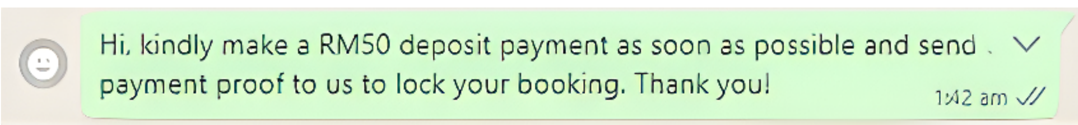


Figure 2.6: Reminder for Late Customers to Pay Deposit

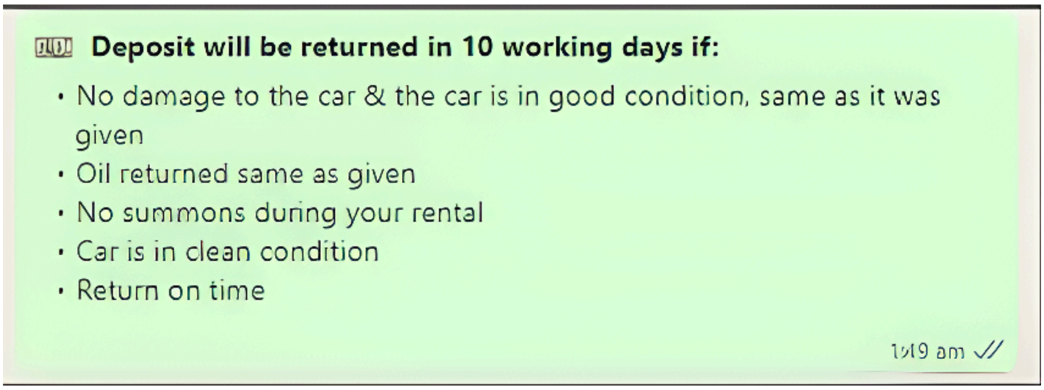


Figure 2.7: Reply Message of Deposit

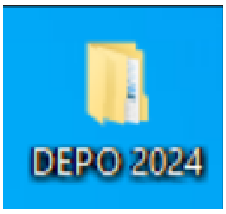


Figure 2.8: Folder for Storing Deposit Receipt by Year

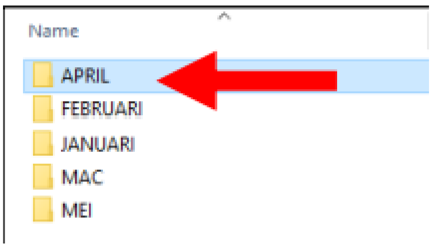


Figure 2.9: File for Storing Deposit by Month

Figure 2.10: Example format file name that follows <year, month, day> <RM50> <name>.

2.3.1.3 Booking Records

The staff will manually record the booking forms on the booking whiteboard to ensure that all bookings are immediately visible and can be easily tracked. They will also use the Drive Wahdah website and a spreadsheet to maintain daily sales reports and booking records, which helps to organize the data systematically, prevent errors , and provide a reliable reference for future verification and reporting purposes.

For the booking whiteboard, each successful booking is recorded in a clearly defined space on the board. The customer's information is entered following a specific format in which the customer's name is written first, followed by the pickup time of the car, the return time and the rental data range. Different colours are used on the board to distinguish types of bookings and users. Black represents staff or UTM students, blue indicates external bookings, green is for Hasta staff and red marks cars that are unavailable for rent. This system allows staff to quickly view which cars are booked, by whom, and during what times, helping to organize daily operations efficiently and prevent scheduling conflicts.

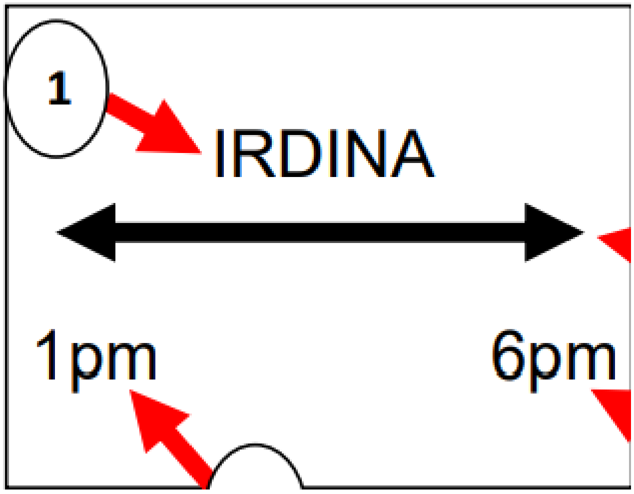


Figure 2.11: The format for writing on the whiteboard

For the Drive Wahdah website, staff are responsible for adding the rental form and filling in all the required information using the booking form and the receipt provided by the customer. This includes recording the customer's personal details such as name, email address, identification number, residential address and mobile number. In addition, the staff will enter the pickup and return information, vehicle details and deposit details written on the note. The pricing section of the form clearly indicates the payment status, specifying whether the payment has been made or remains unpaid. If the customer pays the full rental amount, staff will record the payment date and method of payment, ensuring that all financial transactions are properly documented. Once all the information is completed and verified, the staff will submit the form , thereby creating a complete and accurate record of the rental transaction for future reference, reporting and audit purposes.

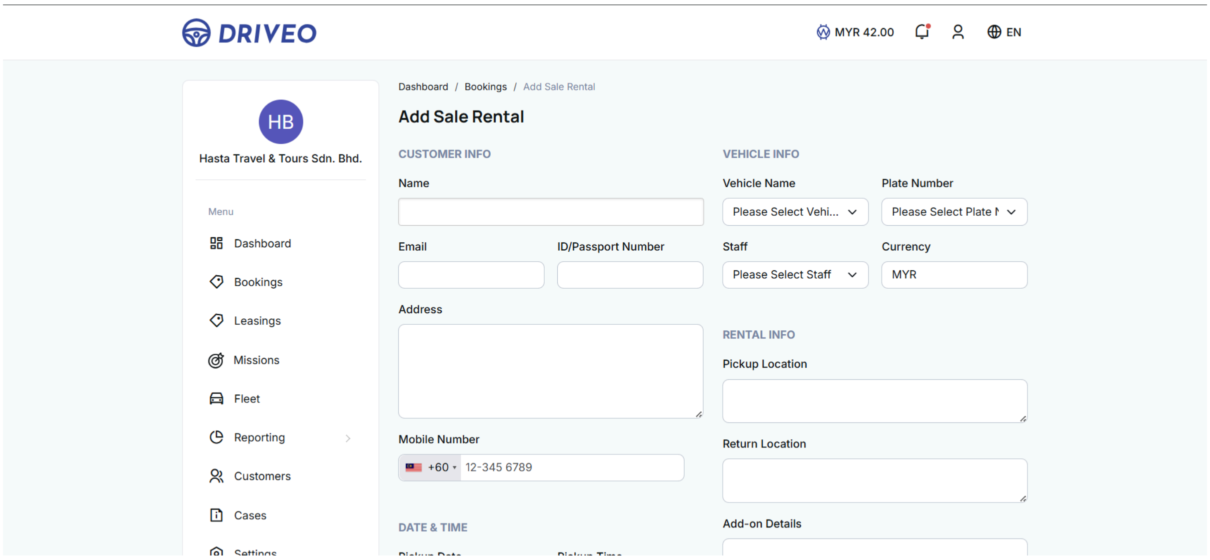


Figure 2.12: Rental Form of Drive Wahdah Website

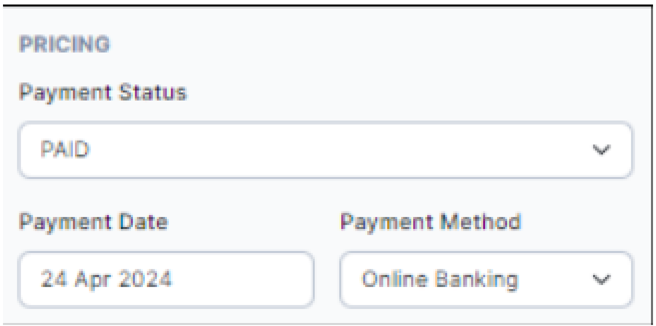


Figure 2.13: Pricing section when the payment status is paid

For the spreadsheet that contains daily sales reports, staff are responsible to fill in the customer's information into the spreadsheet. The process begins by selecting the specific sheet tab at the bottom that corresponds to the actual date of rental. For example, if a customer books a car on 1/4/2025 for the date 2/4/2025, the staff needs to select the tab labeled “2” to record the booking correctly. Then staff must transfer information from the customer's booking form into the “Customer Details” section, including name, bank account and destination. In the “Rental details” section, the staff need to record the destination, the car plate of the rental car and the payment in either the “Cash” or “Transfer" column depending on the method used. Staff must record the deposit amount in the “IN” section and place the value under either the “Cash” or “Transfer" column based on the customer's payment method. They must also include the exact date the payment was received. The “OUT” column is for the deduction part such as deductions for fuel, late returns, or other charges as these are only filled in after the customer returns the vehicle. Lastly , the staff needs to enter their name in the “Key In Data” column. If the person keying in the data is different from the person handing over the keys, both names should be written using a slash (/).

DSR APRIL 2024

4 April

NO	CUSTOMER DETAILS				RENTAL DETAILS									TOTAL (RM)
	NAME	BANK ACCOUNT			DESTINATION	NO PLATE	PAYMENT			PICKUP		RETURN		
		ACC NO	BANK NAME	OWNER'S NAME			CASH (RM)	TRANSFER (RM)	PAYMENT DATE	DATE	TIME	DATE	TIME	
1	Muhammad Azhwan Bin Azhar	151520180686	Maybank	Muhammad Azhwan Bin Azhar	JB	VC6522	0	60	4/4/24	4/4/24	3.55pm			60
2	MUHAMMAD ALIF IMRAN	162405317946	Maybank	Muhammad Alif	jb	NDF9903	0	75	04/04/2024	4/4/24	3.19pm			75
3	FITRI NUL IMAN BIN MOHD RAIS	7631337752	CIMB Bank	FITRI NUL IMAN	jb	JQU1957	0	60	25/3/24	04/04/24	3.11pm			60
4	Mohamad Aqil Bin Mohd Sukri	158275109894	Maybank	Mohamad Aqil B	jb	NDD7803	0	80	04/04/2024	4/4/24	3.45pm			80
5	Ellham Firdausi bin Md Ariff	164490414887	Maybank	Ellham Firdausi	JB	JSM6249	0	70	4/4/24	4/4/2024	5.08 pm			70
6	MUHAMMAD ADZIM ARIFFI	7642357384	CIMB Bank	Muhammad Adz	JB	JSW9545	0	54	03/04/2024	4/4/2024	9.37AM			54
7														0

Figure 2.14: Spreadsheet Daily Sales Reports

IN		
CASH (RM)	TRANSFER (RM)	DATE IN
	50	29/03/2024
	50	01/04/2024

Figure 2.15: “IN” Section of Deposit Details

DEPOSIT DETAILS							PIC (NAME)	
OUT (DEDUCTION)					TOTAL	REFUND (RM)	GIVE CAR/KEY IN DATA	CHECK CAR RETURN
EXTEND PAID (RM)	EXTEND UNPAID (RM)	FUEL (RM)	LATE RETURN (RM)	DATE OUT				
					0	50	AQIL / ADZIM	NAZIFAH
					0	50	AQIL / ADZIM	NAZIFAH

Figure 2.16: "Out" Section of Deposit Details

2.3.2 Pick-up Process

The customer can choose to arrange their car pickup either online or offline, depending on their preference and convenience. Both methods ensure that the customer receives the car and car keys once all necessary steps have been completed.

2.3.2.1 Online Car Pick-Up Process

Before the pick-up process, the staff must ensure that the customer has made the full payment. If the full payment is made by the customer, the staff needs to update the payment status to "Paid" on the Drive Wahdah system. Then, the staff must download both the agreement form and the inspection form in PDF format from the Drive Wahdah system. These forms must be sent to the customer at least 30 minutes before the pick-up time. The staff should also send the procedure of picking up and returning the car to the customer.

For the pick-up process, the customer must sign the agreement form and inspection form and send it back to the Hasta staff. The customer must also ensure the full payment is completed and provide the payment receipt to the staff. After that, the customer can collect the car key, which is placed under the driver's seat. The customer must take photos of the mileage, oil bar, and all four sides of the car.

15 minutes before the pick-up time, the staff will remind the customer about the required documents which consist of the signed agreement form and inspection form, the photos of the car, and the full payment. After receiving all required documents, the staff will then print all the signed agreement and inspection forms, and the PDF copies of the customer's driving licence, IC, and matric card, and store them as hardcopy in the file. Then, the staff must send the customer the car return conditions, which include the rules for oil level, cleanliness, damage, return time, and key return procedure and the place to return the keys.

Lastly, the staff will check the inspection form submitted by the customer. Once the customer sends the four-side photos of the car, the staff must upload the pickup date and time into the spreadsheets.

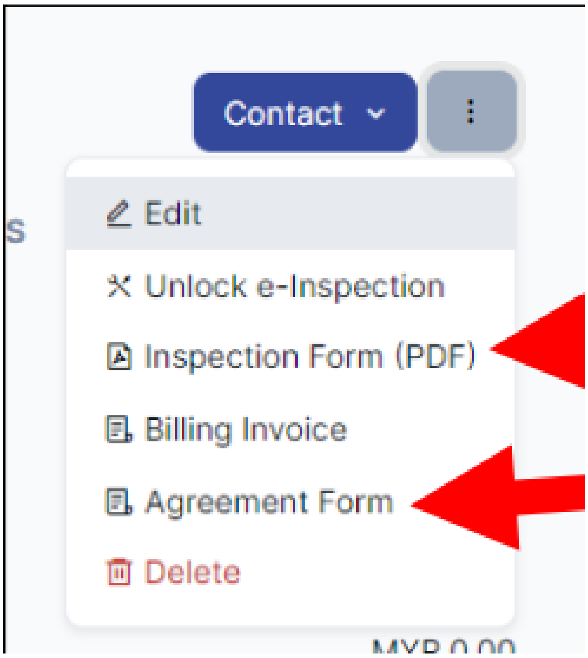


Figure 2.17: The Agreement Form and Inspection Form in the Drive Wahdah System

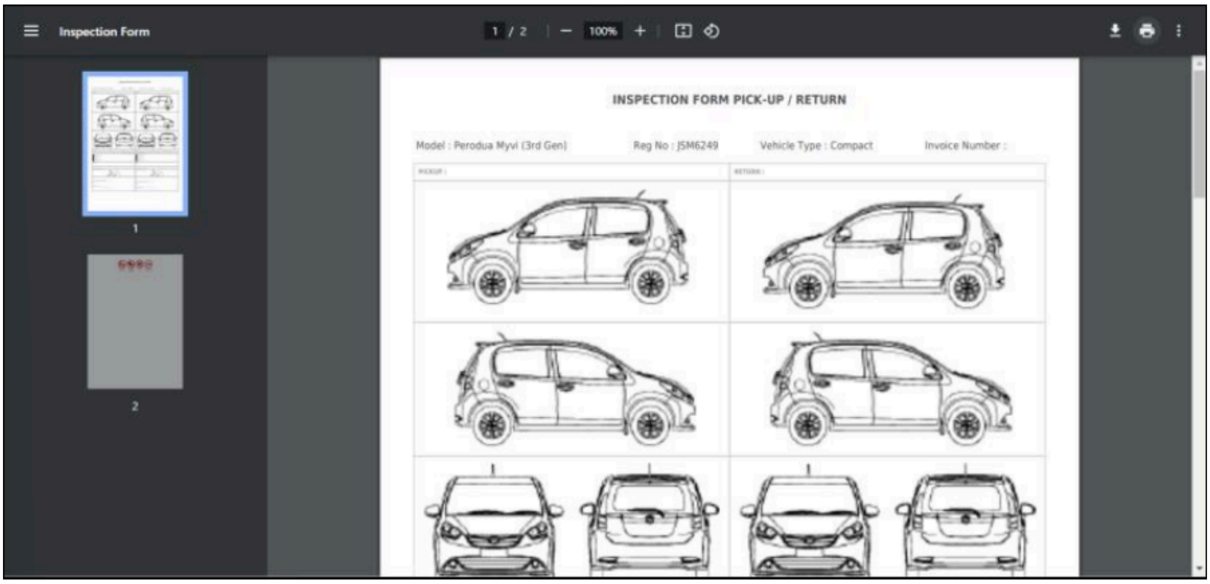


Figure 2.18: Inspection Form

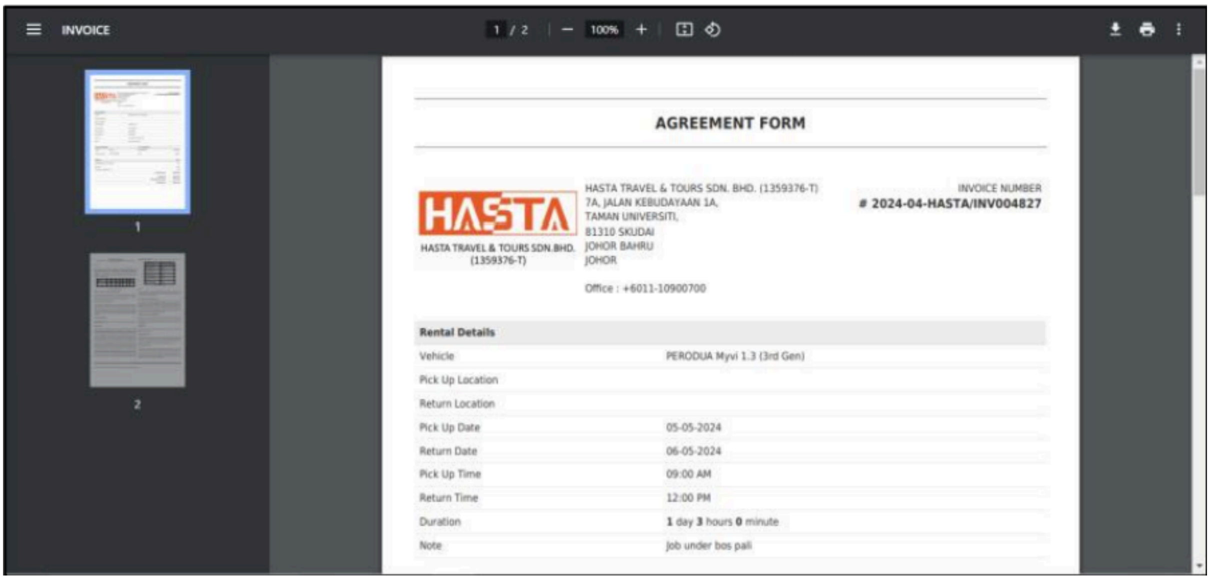


Figure 2.19: Agreement Form

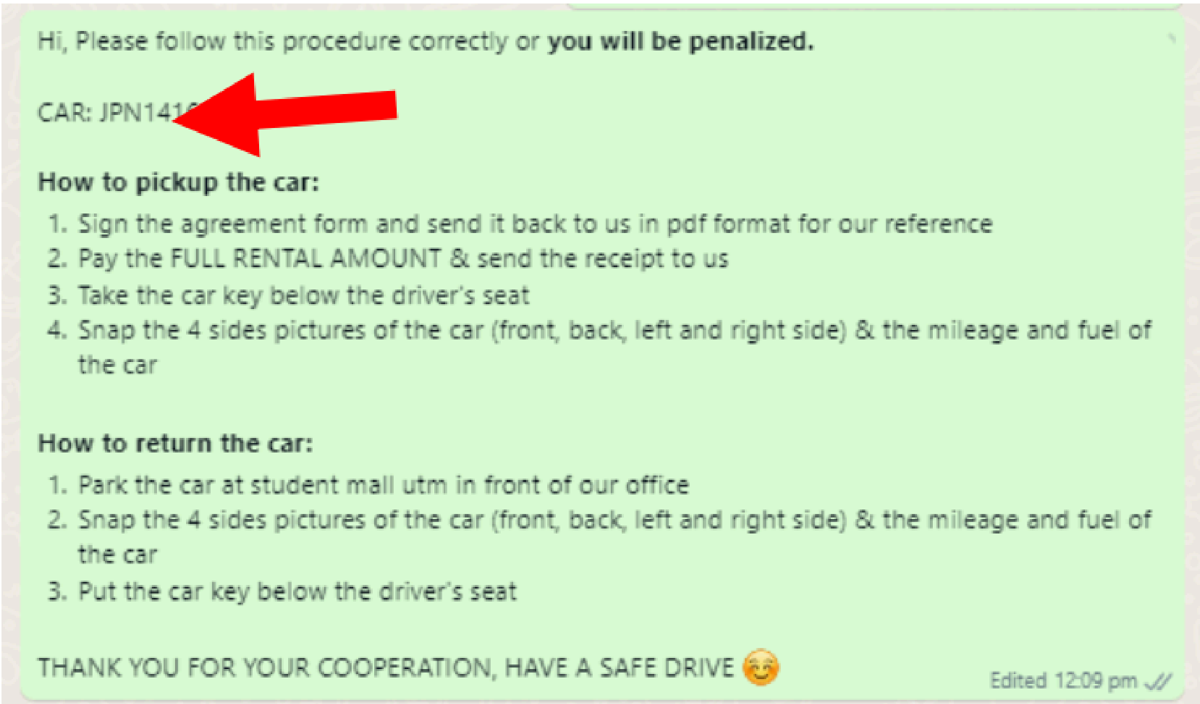


Figure 2.20: Procedure of Pick-up and Return Car

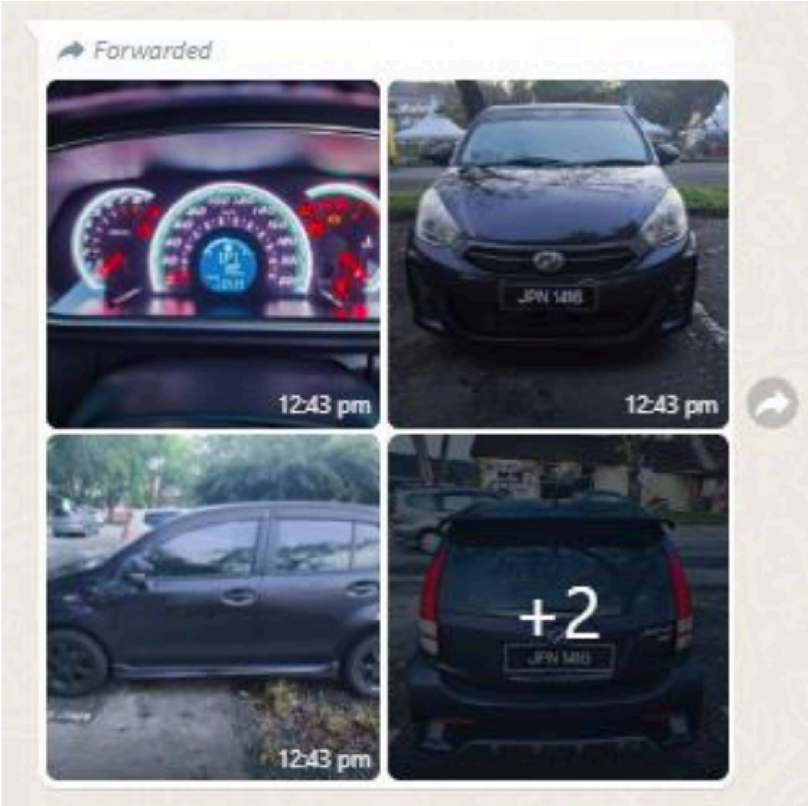


Figure 2.21: The Four-Side Photos of Car that Sent by Customer

Hi. Picture given is the condition of the car before being rented.

Here are some things to pay attention when renting cars at HASTA:-

Oil
Make sure it is returned the same as it was given. If you overfill, no refund will be given.

Deposit will be returned if:

- No damage to the car & the car is in good condition, same as it was given
- Oil returned same as given
- No summons during your rental
- Car is in clean condition
- Return on time

Return Key

- Put the key in the silver Dropbox in front of the store
- Please take a picture of the oil bar, and the car that has been parked in the front of the store.

Penalty will be imposed if:

- The car is returned late (RM30/hour)
- Oil is less than it was given (RM10/bar)

Thank you for using the Hasta Car Rental service

If there are any complaints related to the car or staff, you can email hastatraveltours@gmail.com

12:50 pm

Figure 2.22: Car Returned Condition

2.3.2.2 Physical Car Pick-up Process

Just like the online pick-up process, the staff must first ensure that the customer has made the full payment before the pick-up. If the full payment is made by the customer, the staff needs to update the payment status to “Paid” on the Drive Wahdah system. Next, the staff needs to print the agreement form, the inspection form and the PDF documents that contain the customer’s driving licence, IC, and matric card as hardcopies.

Then, the staff will provide these forms to the customer and ask them to sign the agreement form and the inspection form physically. After signing, the customer must take photos of the mileage, oil bar, and all four sides of the car. Once the customer has taken and submitted all the required photos, the staff may hand over the car key to the customer.

After giving the key, the staff must send the customer the car return conditions, which include the rules for oil level, cleanliness, damage, return time, and key return procedure. Finally, the staff must upload the pickup date and time into the spreadsheet immediately after handing the key to the customer.

HASTA TRAVEL & TOURS SDN.BHD.
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JOHOR

Office : +6013-7104487

2023-08-HASTA/INV004072

Agreement Form

Rental Details	
Vehicle	Perodua Axia 1.0 (A)
Pick Up Location	STUDENT MALL, UTM JB
Return Location	STUDENT MALL, UTM JB
Pick Up Date	01-08-2023
Return Date	02-08-2023
Pick Up Time	01:00 PM
Return Time	01:00 PM
Duration	**1** day **0** hour **0** minute
Note	DEPOSIT RM50.00 (PAID - 1/8)

Customer Details	
Name	NUR SYAFIQAH BT ALI SABRI
Mobile Number	+601139770747

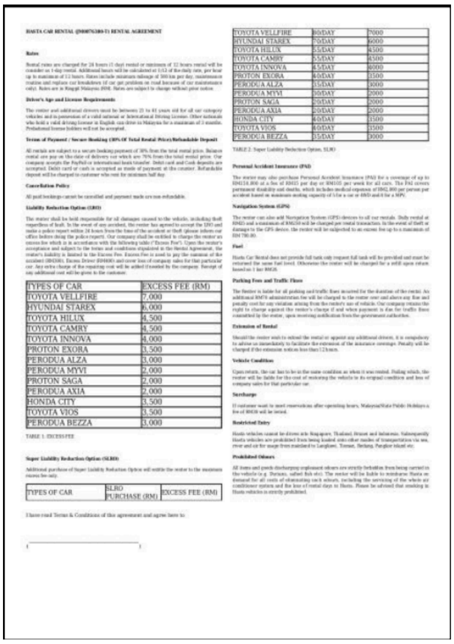
Car Information	
Plate Number	VC 6493
Color	SILVER

Vehicle	**Price**
Perodua Axia 1.0 (A)	90.00
Discount	- 0.00
Total Amount Before Tax	90.00

Total Amount: MYR 90.00

Rounding: MYR 0.00

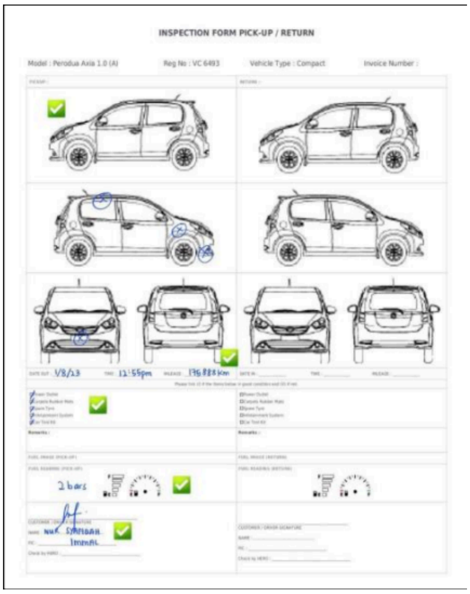
Figure 2.23: Front Page of Agreement Form



TYPES OF CAR	EXCESS FEE (RM)
TOYOTA VELLFIRE	7,000
HYUNDAI STAREX	6,000
TOYOTA HILUX	4,500
TOYOTA CAMRY	4,500
TOYOTA INNOVA	4,000
PROTON EXORA	3,500
PERODUA ALZA	3,000
PERODUA MYVI	2,000
PROTON SAGA	2,000
PERODUA AXIA	2,000
HONDA CITY	3,500
TOYOTA VIOS	3,500
PERODUA BEZZA	3,000

TYPES OF CAR	SLRO PURCHASE (RM)	EXCESS FEE (RM)
TOYOTA VELLFIRE	80/DAY	7000
HYUNDAI STAREX	70/DAY	6000
TOYOTA HILUX	55/DAY	4500
TOYOTA CAMRY	55/DAY	4500
TOYOTA INNOVA	45/DAY	4000
PROTON EXORA	40/DAY	3500
PERODUA ALZA	35/DAY	3000
PERODUA MYVI	30/DAY	2000
PROTON SAGA	20/DAY	2000
PERODUA AXIA	20/DAY	2000
HONDA CITY	40/DAY	3500
TOYOTA VIOS	40/DAY	3500
PERODUA BEZZA	35/DAY	3000

Figure 2.24: Back Page of Agreement Form



INSPECTION FORM PICK-UP / RETURN

Model : Perodua Axia 1.0 (A) Reg No : VC 6493 Vehicle Type : Compact Invoice Number :

Figure 2.25: Inspection Form

2.3.3 Return Process

When returning the car, the customer must ensure it is returned on time, in a clean condition, with the oil level the same as when it was given, and no summons issued during the rental period. The customer should place the key in the silver Dropbox in front of the store and take a picture of the oil bar and the car that has been parked in front of the store which is on all four sides of the car, along with the mileage and fuel bar. If the car is returned late or the oil level is lower than given, penalties will apply. The customer will be fined RM30 per hour if returned late and RM10 per bar if the oil is less than it was given. Customers should record the date, time, mileage, and fuel level of the car, and fill in their name in the PIC section of the inspection form. The staff should enter 0 in the Extend Paid, Extend Unpaid, Fuel, and Late Return fields if the car is returned in the same condition as rented. After completing the return, staff should notify the customer that the deposit will be returned within 10 days.

If the customer late returns the car, staff must record the delay duration in the REMARKS section and note the lateness on the front page of the Agreement Form. The delay is counted after 30 minutes past the scheduled return time. If the oil level is not the same as when given, the staff should record the amount of oil missing in REMARKS and make a note on the front page of the Agreement Form. In the case of car damage, such as dents, multiple scratches, or broken parts, staff must inform the customer by comparing the car with the pre-rental photos, mark the damaged areas on the photos, label the type of damage, and also record the details on the front page of the Agreement Form.

If a customer requests to extend the car rental, staff should record the extension fee per hour according to the type of car in the “Extend” column. For example, RM35 per 1 hour extension for a Myvi. Extensions can be marked as “Extend Unpaid” if the fee is deducted from the customer’s deposit, or “Extend Paid” if the customer has paid the fee directly. If the customer returns the car late, staff should record the late fee per hour according to the car type in the “Late Return” column. For example, RM30 per 1 hour delay for an Axia. If the car is returned with less fuel than given, staff should record RM10 per missing fuel bar in the “Fuel” column. For example, RM10 for 1 missing fuel bar.

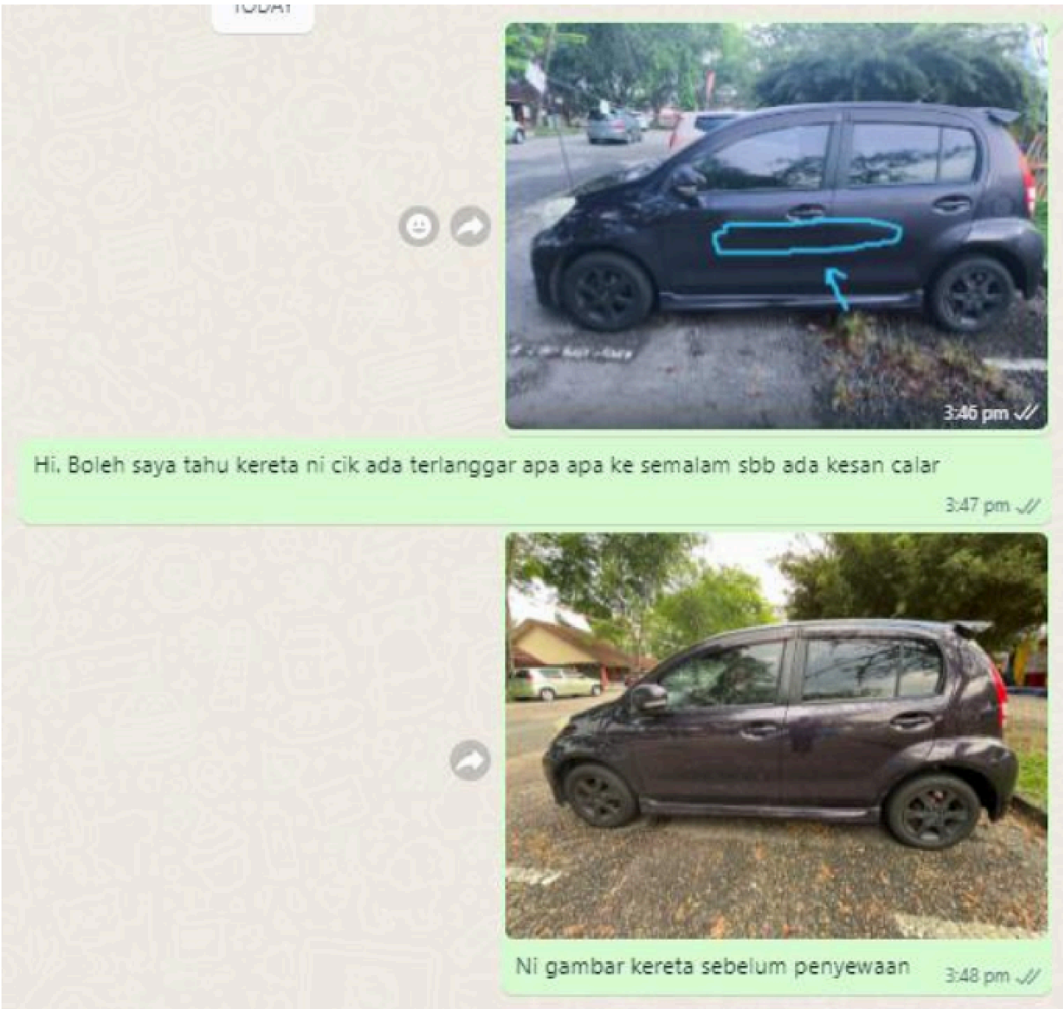


Figure 2.26: Comparison with Pre-Rental Photos of Car

2.3.4 Loyalty Card and Reward

The customer's loyalty card and rewards will be recorded in Excel. To get a loyalty card, customers must rent a car for 12 hours, regardless of car type, which earns them the first stamp. To get a stamp, the customer must rent a car for 9 hours. After issuing the loyalty card, staff should record the customer's name in Excel according to the serial number on the card, mark the first box with a (/) and note the date and the name of the person-in-charge (PIC). If the customer rents for 9 hours again and reaches a box that qualifies for a discount, staff should mark the "Claimed" button to indicate the reward has been used.

CARD NO	CUSTOMER DETAILS
	NAME
001	

Figure 2.27: Loyalty Card that Need Serial Number and Name of Customer

BOX 1	TARIKH	PIC
☐		
☐		

Figure 2.28: Loyalty Card

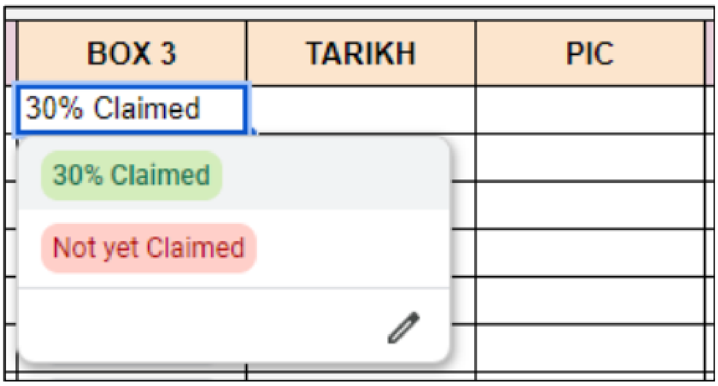


Figure 2.29: Loyalty Card that Can Claim

2.4 Problems Faced by Client (Users) in the Existing System

2.4.1 Issues or Problems with Existing Systems

Hasta Travel’s manual system has a number of serious problems that impact the accuracy, scalability, and operational efficiency of the business. The system is unsustainable as booking volume rises due to its significant reliance on manual Driveo entries, WhatsForm submissions, and WhatsApp messaging.

First of all, the booking procedure is inefficient and incredibly prone to mistakes. The staff must manage several requests at once because all correspondence and confirmations are completed manually via WhatsApp, which raises the possibility of duplicate bookings and miscommunication. An effective synchronization of hourly and daily rental schedules is impossible in the absence of a real-time vehicle tracking system.

Second, the customer experience is adversely affected by the absence of automation and a digital user interface design. In contrast to Wahdah, SOCAR and GoCar systems, Hasta's website does not let users make booking or payment directly online. Instead, it only shows basic text information. Customers are forced to use manual communication, which reduces satisfaction and delays response times.

Third, the current booking system requires customers to provide numerous documents, such as IC, driver’s license, and matric card, which makes the process lengthy and complicated. As a result, customers have to go through many steps to complete a booking, leading to potential frustration and dissatisfaction with the service.

Fourth, scheduling and geographic limitations make services even less accessible. The current booking system restricts the pickup location to only Student Mall and the drop-off location to areas around UTM. This limitation prevents students who are outside these locations from making a booking, which reduces the number of potential customers.

Additionally, the current system enforces a “cool-down” period following every rental, meaning that if a car is returned the next morning, the system blocks the entire day from new bookings. This limitation requires staff to manually check car availability, which slows down the booking process and can lead to reduced efficiency and customer dissatisfaction.

Obviously, these limitations make it abundantly evident that Hasta's manual workflow is unable to meet customer expectations or operational demands in the modern digital world.

3. The Proposed System

This section describes the proposed To-Be system for the Hasta Vehicle Rental Booking System. It outlines the purpose and goals of the system, explains how the system will operate compared to the current manual process, and defines its scope and boundaries. The section also presents the system's mission statement, key objectives, main functionalities, involved user roles, and the technologies proposed for designing and developing the system.

3.1 Mission Statement

The mission of the Hasta Vehicle Rental Booking System is to provide a centralized and efficient web-based platform that automates the vehicle rental booking process for Hasta Travel & Tours. The system is intended to replace the existing manual workflow by improving operational efficiency, reducing human errors, and enhancing transparency in booking activities, fleet management, and customer interactions.

3.2 Overview of the To-Be System

The proposed Hasta Vehicle Rental Booking System is a role-based web application designed to manage the full vehicle rental lifecycle in a structured and organized manner. The system supports two main user groups: customers, which include UTM students and staff, and clients, consisting of Hasta staff and administrators. Customers can register and log in to the system, browse available vehicles, make rental bookings, upload payment receipts, submit vehicle inspection images, and provide reviews after completing the rental period. Meanwhile, Hasta staff and administrators use the system to manage fleet information, review and approve bookings, verify customer submissions, monitor rewards usage, and generate administrative reports. By digitalizing these processes, the system ensures accurate record-keeping, improves traceability, and increases overall operational efficiency.

3.3 Mission Objectives

The proposed database system has been designed with a set of clearly defined objectives to address the shortcomings of the current car rental booking process and fulfill the requirements of both clients and staff.

1. To automate vehicle rental booking operations, reducing reliance on manual processes.
2. To minimize errors associated with manual record-keeping and data entry.
3. To improve overall service efficiency for both customers and staff.

4. To provide real-time vehicle availability information to support faster decision-making.
5. To establish a structured booking approval workflow that ensures consistency and accountability.
6. To organize and manage rental documents in a centralized and accessible manner.
7. To support administrative decision-making through comprehensive reporting features.
8. To enhance customer satisfaction by offering a clear and guided rental process.

3.4 Scope

Our proposed system is a usable and easy to maintain system which is implemented and designed to manage the car rental operations of Hasta Travels. There are three types of system users that are categorized into customer, staff and admin. In the Hasta Vehicle Rental Booking System, there are several modules including user management, car management, booking management, payment and penalty handling, loyalty rewards and admin dashboard modules.

The system can be accessed through the general website via any search engine. Users can either log into their existing account or sign up for a new account. As for staff and administrators, they are only permitted to log into the system. Staff accounts can only be created through a page that is exclusively accessible by administrators.

3.4.1 User Management

As for customers, upon logging into the system, they will be redirected to the customer interface. The customer interface consists of a main dashboard which redirects to multiple different pages including the home, vehicle, booking, vouchers and help pages. Customers also have their own profile page which contains all of their personal information, including documents such as their identity card or passport, matriculation card and driving licence pictures. Before proceeding to make a booking, customers are required to complete their profile and have their documents manually verified by staff.

If a user is blacklisted because of outstanding payments, they will still be able to log in and view the car models and details of the car. However, they would be restricted from accessing other system features. Once the blacklisted user wants to rent a car, they will be redirected to a payment page to clear their outstanding payments.

3.4.2 Car Management

In the vehicles page, customers are able to view all car models handled under the company along with their respective details such as price, colour, engine, power and number of seats. Real-time availability is also shown via an interactive calendar that highlights when a car is booked and enforces a 1-hour cooling period between bookings. Customers may proceed to make a booking directly from this page.

3.4.3 Booking Management

Other than that, customers can perform a rental booking which involves selecting a vehicle and a preferred pick-up and return location, as well as the date and time. If a customer chooses a location other than Student Mall for pick-up or return, an extra fee will be charged. Customers are also given the option to input a voucher code obtained from the vouchers page. Following that, the booking fee is automatically calculated based on the rental period, pick-up point, a deposit of RM50 per booking and any voucher applied. A full payment breakdown will be displayed and customers may proceed to pay the total rental amount via QR payment. Once payment has been made, the customer will need to wait for staff verification and approval in order to activate the booking, after which a confirmation notification will be sent through their email and on the website.

Before picking up the car, customers are required to take photos of the car, fill in the current fuel level and upload the documents along with the agreement form. The system will then automatically start the countdown once the uploading process has been completed. A few minutes before the rental period ends, customers are required to upload after-rental photos along with the inspection form. Customers are also given the option to leave feedback and ratings regarding their overall rental experience, whether about the service or the car. If a customer wishes to cancel the booking, it must be done at least 24 hours before the pick-up time in order to receive a full refund including the deposit. Cancellations made within 24 hours of the pick-up time will not be eligible for a refund.

3.4.4 Payment and Penalty Handling

The system will also handle payment and penalty tracking. The system supports QR payment, and payment receipts must be uploaded for verification. Late returns are subject to penalties, as users are given a 30-minute grace period after the return time. After this period, an additional fine of RM50 will be charged for each hour of overdue return, regardless of whether the vehicle is a car or a motorcycle.

3.4.5 Rewards

The vouchers page will display all available vouchers that can be redeemed by customers based on their loyalty stamps. Once a voucher has been redeemed, it must be used within a specified time limit before it expires. Vouchers such as a 12-hour free booking can be applied during the booking process to reduce the total rental cost.

3.4.6 Admin Dashboard

The admin dashboard displays important activity and information needed to be reviewed, such as total revenue, active rentals, pending tasks, quick links and an action center. The manager dashboard contains more visualised graphs that show performance trends, top rented models, customer analytics and fleet and booking status. Both managers and administrators are able to verify user documents, approve or reject customer bookings, update car availability and manage blacklists while tracking payments.

The fleet page allows staff to add or edit vehicle information, including rental prices, vehicle documentation and other relevant details. The bookings page displays all bookings recorded in the system along with all relevant information compiled throughout the booking process, including receipts, photos, inspection forms and reviews.

The reporting page displays revenue and performance data based on preferences set by the administrator. The data can be filtered by yearly, monthly or weekly intervals and adjusted by categories. This page also offers a print feature for documentation purposes.

3.4.7 Out of Scope

The system does not handle vehicle insurance claims or policy renewals. All insurance-related matters would remain under Hasta Travels' external insurance provider to avoid integration with third-party insurance APIs and confidential legal data. In addition, the system does not include real-time vehicle tracking using GPS. Vehicle location monitoring and live tracking are not implemented to avoid additional hardware requirements and system complexity. Lastly, although the system accepts payment receipts, it does not connect directly to bank systems. All payment verifications are done manually by staff to avoid the complexity of financial API compliance.

3.5 System Boundaries with Use Case Diagram

The system boundary defines the interactions between users and the Hasta Vehicle Rental Booking System while excluding external entities.

Customers interact with the system through use cases such as user register, user log in, browse fleet, upload document, edit customer profile,pickup vehicle, view booking history, return vehicle, cancel booking, booking vehicle, payment and view all vouchers.

Hasta staff interact with the system to manage bookings, approve rentals, update vehicle status, verify documents and view customer profiles.

Hasta administrators have additional privileges, including managing staff accounts and accessing reporting functionalities.

External entities such as QR payment services and manual vehicle inspection activities remain outside the system boundary, with only their outcomes recorded within the system. These interactions are represented in the Use Case Diagram, which clearly distinguishes user roles and system responsibilities.

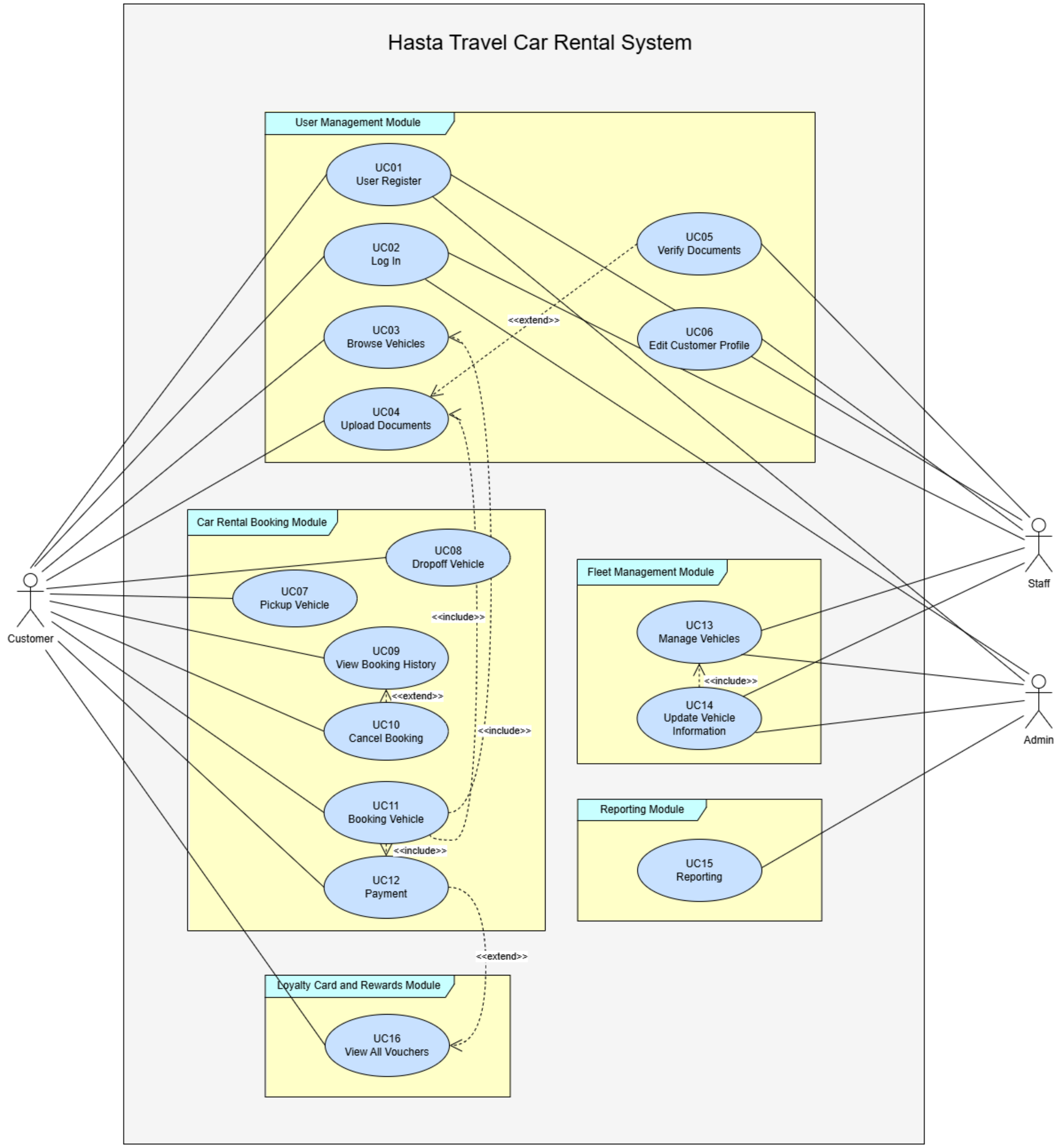


Figure 3.5.1 : Use Case Diagram

3.6 Proposed Technology to Be Used

Components	Proposed Technology
Frontend	HTML, CSS & JavaScript
Backend	PHP
Hosting	Laragon
Database	MySQL
Security	Laravel's Built-in Security Features, user authentication (staff/customer), blacklist access control

Table 3.6 : Proposed Technologies to be used

The proposed system will be designed and developed using the Laravel web application framework, built on PHP as the backend language. Laravel is selected due to its structured development approach, scalability, and built-in security features. The framework follows the Model–View–Controller (MVC) architectural pattern, allowing a clear separation between data management, application logic, and user interface components. The frontend of the system will be developed using HTML, CSS and JavaScript, while Laravel's Blade templating engine will be used to render dynamic user interfaces. MySQL will be used as the relational database management system to store booking records, vehicle details, user information, and uploaded documents. For the development environment, Laragon will be used to host and run the system locally during development. In terms of security, the system will utilize Laravel's built-in security features along with user authentication for both staff and customers, as well as blacklist access control to restrict unauthorized access. This technology stack ensures maintainability, security, and efficient development for a web-based vehicle rental booking system.

4. Glossary

The following table 4.1 defines the technical terms, acronyms, and domain-specific terminology used throughout this proposal.

Term / Acronym	Definition
Blade Templating Engine	A simple yet powerful templating engine provided with Laravel. It allows the application to render dynamic user interfaces and display data passed from the controller.
Cool-down Period	A specific duration enforced by the existing system after a vehicle is returned, preventing it from being booked immediately to allow for maintenance or turnover.
Drive Wahdah	The existing third-party system currently used by Hasta Travel staff for internal management of booking records and payment status.
Laravel	A free, open-source PHP web framework intended for the development of web applications following the Model–View–Controller (MVC) architectural pattern.
Model–View–Controller (MVC)	A software architectural pattern that divides the related program logic into three interconnected elements: the Model (data), the View (user interface), and the Controller (processes/logic).
MySQL	An open-source relational database management system (RDBMS) that will be used to store and manage the data for the proposed system.

Role-based Access Control	A method of restricting network access based on the roles of individual users within an enterprise. In this system, it distinguishes between "Customers" and "Admins/Staff."
Stakeholder	An individual, group, or organization that has an interest in the project and can affect or be affected by the system's development.
Use Case Diagram	A type of behavioral diagram defined by the Unified Modeling Language (UML) used to illustrate the interactions between the system and its users (actors).
UTM	Universiti Teknologi Malaysia. The primary location and customer base (students/staff) for the vehicle rental service.
WhatsForm	An online form builder tool currently used by the client to collect booking details from customers and convert them into WhatsApp messages.
WhatsApp	A messaging application used as the primary communication channel in the existing manual system for booking confirmations and customer support.

Table 4.1 Glossary for acronym/term & definition

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